SARA TANKARD

Address: 404 Water Street Apt 1902, Baltimore, MD 21202
Phone: (443) 621 1240
E-mail: sara.tankard@gmail.com

EDUCATION

Johns Hopkins University, Baltimore, MD May 2012
GPA: 3.20
Double Major: Engineering Mechanics & Neuroscience **Concentration:** Biomechanics

EXPERIENCE

Software Engineer Intern, Jacó Costa Rica *Feb 2013- Feb 2014*
GeckoTech S.A

- Lead programmer for developing a remote control service for local sports bar
  - Custom programmed KP-900 Universal remote control to control 8 TVs and 8 satellite feeds
  - Developed a Java web service (Java, JSP, HTML, CSS, XML) "online remote control" to monitor and change the state of a matrix switch (via RS-232 serial port), deployed on apache tomcat server
- Entry-level co-programmer for an energy management system to intelligently control air conditioning units (using Z-wave sensors and VERA home automation technology)
  - Debugged, modified, and added code (Lua) responsible for remotely turning off AC in order to improve program efficiency and allow code compatibility with new technologies
  - Managed databases in SQL for analysis of events (door open, AC turned off, etc) order to both locate failures in devices/code and generate client reports on home events and energy savings
  - Interpreted detailed log events reported remotely from VERA servers via SSH to debug code
- On-site instructor for hotel/restaurant point-of-sale software to teach restaurant staff how to utilize software and for owners to maximize back-end data management (such as sales statistics)

Research Coordinator, Johns Hopkins University, Baltimore, MD *June 2012- Feb 2013*
Dept. of Physical Medicine Rehabilitation, Human Brain & Physiology Lab

- Used MATLAB to develop algorithms for data analysis and graphic presentation in the investigation of transcranial direct current stimulation (tDCS) on working memory in healthy human subjects
- Recruited and coordinated human volunteer experiments by answering e-mails and phone calls as well as being responsible for coordinating inter-departmental events and other organizational tasks

Student Researcher, Johns Hopkins Bayview Medical Center, Baltimore MD *Mar 2010- June 2011*
Orthopaedics Lab

- Investigated the effect screw insertion torque on pullout strength in normal osteopenic and osteoporotic cadaveric human humeri (Tankard et al. 2012 *Journal of Orthopedic Trauma)*
- Developed an algorithm (MATLAB) to find far cortex of human humeri and analyze screw insertion torque patterns

Student Intern, Imperial College London, London, UK *June 2011- Aug 2011*
Charing Cross Hospital, Biodynamics Lab

- Conducted research and created data analysis algorithms using MATLAB to compare walking performance of patients after knee replacement procedures

ACTIVITIES

Volunteer at Esperanza Center for Latino Immigrants Aug 2014
JHU Mechanical Engineering Senior Design Team 2011-2012

- Worked on 4-member team to design, develop, program, actuate, prototype a novel automatic material selection device (AMSD) to route plastic materials to various locations in a plastics factory.
- Awarded best overall design by American Society of Mechanical Engineers (ASME**)**

JHU Varsity Soccer Player 2007-10

LANGUAGES

Java, Javascript, JSP, HTML, CSS, XML, Lua, SQL, MATLAB; Spanish